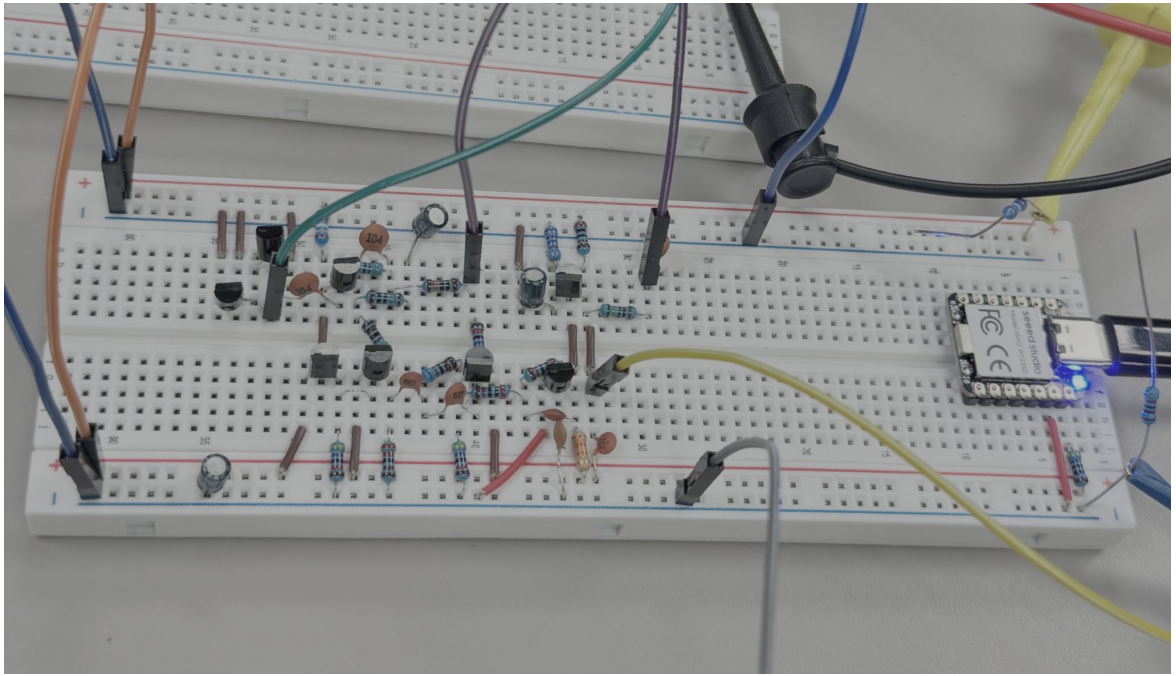
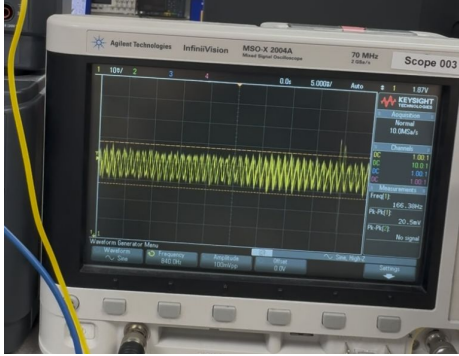
October 21st Progress Report

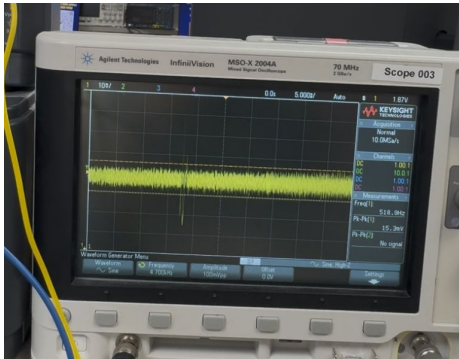
Ethan Ewer

Problem (repeated from last week): Custom Amplifier has a lot of noise









Solution (repeated from last week):

- Use an existing design that meets our requirements
- Add passive pre-amp and post-amp filters

Requirements:

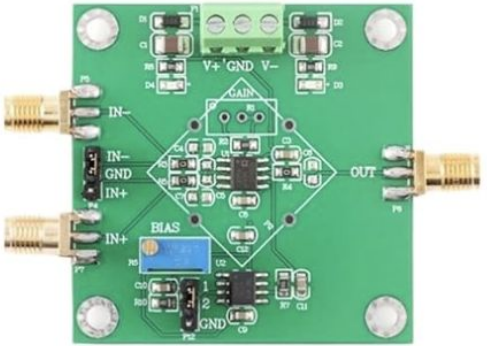
- Input noise under 4.5 nV/√Hz
- Input impedance over 1 MΩ
- 200 V/V gain
- Bandwidth flat up to 2.2 kHz

AD8421 selected in pre-built amplifier

Requirements:

- Input noise under 4.5 nV/√Hz (gets 3.2 nV/√Hz)
- Input impedance over 1 MΩ (gets 30 GΩ)
- 200 V/V gain (configurable from 1-10000 V/V)
- Bandwidth flat up to 2.2 kHz (yes)





Updated BOM

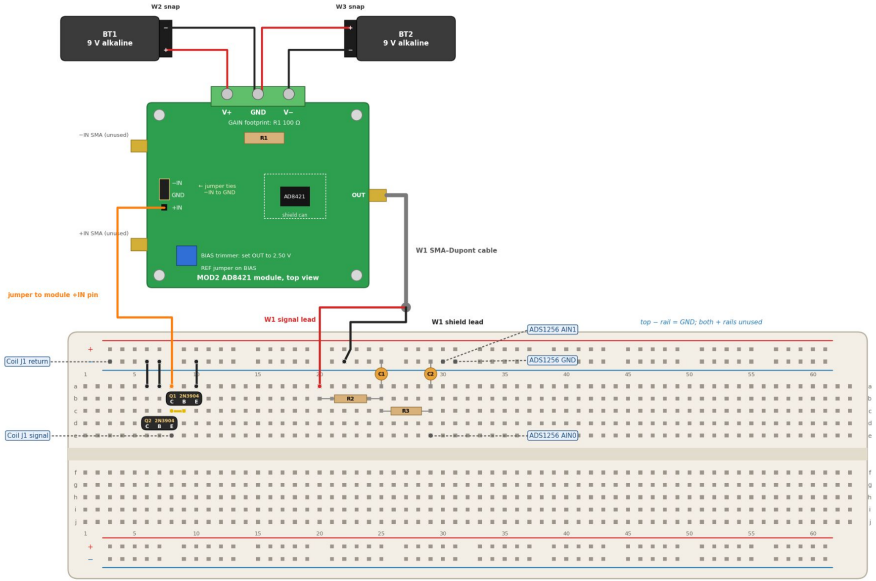
I18 fx

	A	B	C	D	E
2		Item	Cost	Quantity	Total Cost
3		Kapton tape	$9.99	1	$9.99
4		Pico 2	$13.43	1	$13.43
5		ADC1256	$16.99	1	$16.99
6		26AWG Magnet Wire	$30.90	2	$61.80
7		22AWG Magnet Wire	$29.90	1	$29.90
8		Digikey Order for Polarization Board	$59.49	1	$59.49
9		Deionized water	$16.95	1	$16.95
10		AD8421	$28.68	1	$28.68
11		Dupont male coaxial test lead	$9.98	1	$9.98
12		9 V battery snap connector	$5.30	1	$5.30
13		Clip-on ferrite core assortment	$8.99	1	$8.99
14		9 V alkaline battery	$4.99	1	$4.99
15					0
16					0
17					0
18					0
19					0
20					0
21					0
22					0
23					0
24					0
25					0
26					0
27					0
28				Total Used:	$266.49
29					

Updated Design

Receiver assembly — AD8421 module, 9 V batteries, and an 830-point breadboard

Top view. Breadboard holes, parts, and jumpers match spice/receiver.cir node for node. Only the top − rail is used (GND); neither 9 V supply touches the breadboard.

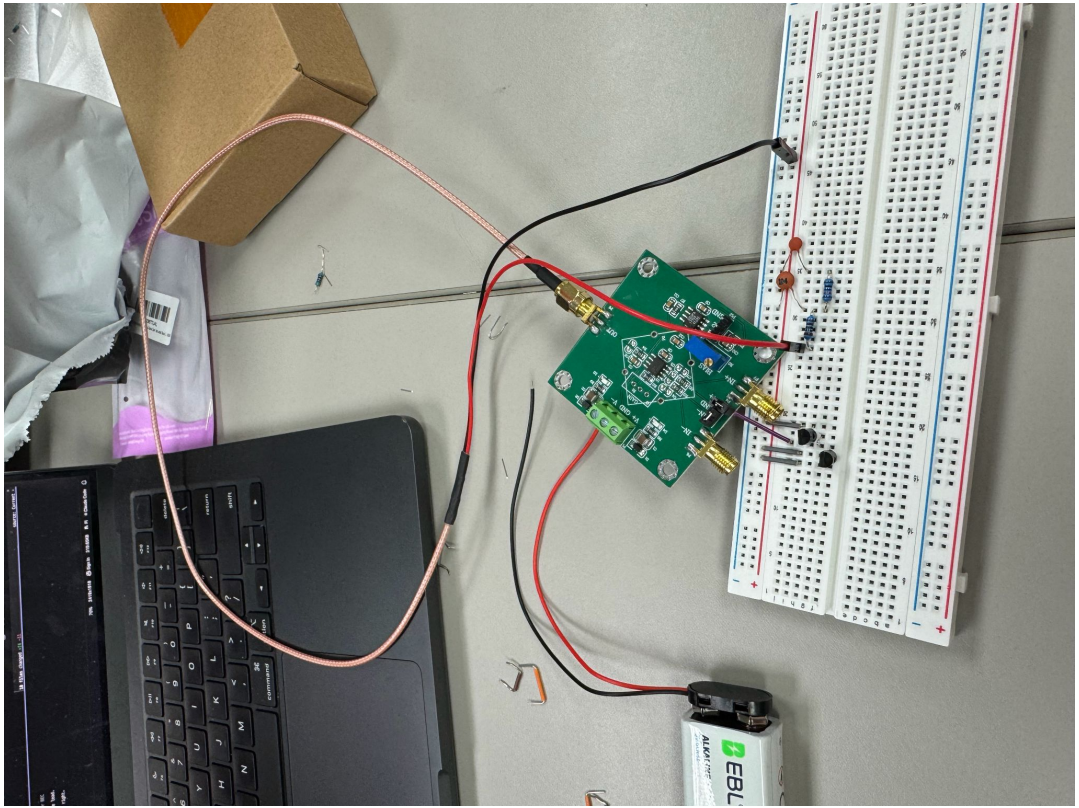


Part values

R1 100 Ω (on module)	R2 1 kΩ	R3 10 kΩ	C1 22 nF	C2 2.2 nF	Q1 2N3904	Q2 2N3904

1. Prepare the module first (assembly.md); confirm the AD8421 marking, remove the input RFI parts between the input jacks and the chip, and solder R1 (100 Ω) into the GAIN footprint.
2. Batteries go only to the module's screw terminal: BT1 + → V+, BT1 − and BT2 + twisted together → GND, BT2 − → V−. Disconnect a battery when idle.
3. W1: plug the SMA end into OUT. Find which Dupont lead reaches the SMA center pin with a meter; that lead goes to a20 and the other (shield) to the top − rail. The shield is the only ground link between the module and the breadboard.
4. The orange jumper (male to female Dupont) runs from a8 to the module's +IN header pin; keep it short. Leave the header jumper on −IN/GND and the REF jumper on BIAS.
5. Q1 and Q2 are 2N3904 clamp diodes. The letters give the lead each hole needs (C, B, E): check the pinout of each kit transistor before inserting it.
6. With J1 shorted, trim the BIAS trimmer until OUT reads 2.50 V; then AIN0 (e29) reads about 2.39 V. Check the module GND terminal to the top − rail reads under 1 Ω.
7. ADS1256: buffer off, PGA 1, 30 kSPS, AIN0 − AIN1. Power it and the XIAO from USB; wire SPI directly: D8→SCLK, D10→DIN, D9←DOUT, D3→CS, D2←DRDY, D4→SYNC/PDWN.
8. Rails are drawn continuous. Every rail hole used is in the left half, so a board that splits its rails at the middle needs no bridge.

In progress: testing new amplifier (it came this morning)



Next Steps

- Wait for ordered amplifier to arrive (estimated arrival by 10/26) and then test it
- Test amplifier with ADC and MCU
- Test end-to-end device

Feedback

